

7 Planar and Non-planar Graphs

A graph is a **planar graph** if it can be drawn in the plane with no edge crossings. [Note: You may need to rearrange the vertices and edges to avoid crossings.]

7.1 Euler and Planarity

We have already seen **Euler's Formula for Planar Graphs**:

$$v - e + r = 1 + \text{number of connected components},$$

where v is the number of vertices, e the number of edges and r the number of regions.

In particular, for a connected graph we have $v - e + r = 2$.

1. Show that the following are planar.
 - a. $K_{2,2}$
 - b. $K_{2,3}$
 - c. $K_{2,4}$
2. Is $K_{2,n}$ planar for every n ?
3. Use Euler's formula to show that K_5 is not planar.
 - a. How many vertices does K_5 have?
 - b. How many edges does K_5 have?
 - c. If K_5 were planar, how many regions would it have?
 - d. Show that K_5 can't have that number of regions, so it must not be planar.
4. Use Euler's formula to show that $K_{3,3}$ is not planar.

7.2 Kuratowski's Theorem

The proof of the following theorem is too involved for this course, but it turns out that all nonplanar graphs contain a "homeomorphic copy" of either K_5 or $K_{3,3}$.

Kuratowski's Theorem: A graph G is nonplanar if and only if it contains a subgraph H that is homeomorphic to either K_5 or $K_{3,3}$.

- two graphs are **homeomorphic** if they can each be obtained from the same graph by a sequence of elementary subdivisions.
 - an **elementary subdivision** of a graph G removes one edge (u, v) and adds a new vertex n and two new edges (u, n) and (v, n) .
5. Create a graph with 5 vertices and 7 edges. Now perform three elementary subdivisions on your graph.
 6. Describe what an elementary subdivision is in simple words a kindergartner could understand.
 7. Explain why two homeomorphic graphs are either both planar or both nonplanar.
 8. Use Kuratowski's Theorem to show that the Petersen graph is nonplanar.

